A.7 Log Book for Capstone Project

Mukesh Patel School of Technology Management & Engineering

SVKM's NMIMS (Deemed-to-be University), Indore Campus

Log Book for Capstone Project

Department: Artificial Intelligence and Data Science

Program: B.Tech in Artificial Intelligence and Data Science

Semester: VII

Academic Year: 2026-27

TITLE OF THE PROJECT: MemGate: An Adaptive Memory Layer for Long-Running LLM Agents

Name of the Faculty Mentor: Dr. Praveen Modi, Assistant Professor

STUDENT DETAILS

	NAME	ROLL NO.	CONTACT
Student 1	Yash Ramchandani	S024	
Student 2	Simar Singh Khanuja	S019	

Name of group members:

Roll No.	Name	Sign (Students)	Grades (By Faculty Mentor)
S024	Yash Ramchandani		
S019	Simar Singh Khanuja		

SAP IDs: Yash Ramchandani (70562300097), Simar Singh Khanuja (70562200088)

Week No: 1

Date of Reporting: 22/07/2026

18/07/2026 to 23/07/2026

Work carried out

- Identified the problem: LLM agents re-send their whole history on every turn, so cost and latency grow and accuracy falls well before the context limit.
- Literature survey of memory for LLM agents (MemGPT, MemoryBank, Generative Agents, A-MEM) and of prompt and KV-cache compression (LLMLingua, Gist tokens, AutoCompressors, StreamingLLM, H2O).
- Selected MemGPT (Packer et al., 2023) as the base paper.
- Drafted the problem statement, three objectives, scope and the semester Gantt chart.

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Week No: 2

Date of Reporting: 29/07/2026

25/07/2026 to 30/07/2026

Work carried out

- Presented Review 1: introduction, problem statement, objectives, scope, proposed system and architecture, implementation workflow, tools and technology, feasibility and Gantt chart.
- Presented a comparison of 10 papers (dataset, technique, achievements, limitations, gap addressed).
- Noted the panel's questions: is the project needed now that context windows are large, is it feasible, is a tiered memory still novel, and what is the end product.

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Week No: 3

Date of Reporting: 05/08/2026

01/08/2026 to 06/08/2026

Work carried out

- Studied the long-conversation benchmarks LoCoMo, LongMemEval and LongBench and chose LoCoMo as the main benchmark (marked evidence turns allow a retrieval-level metric).
- Chose the stack: Python, tiktoken for exact token counts, all-MiniLM-L6-v2 for embeddings, Qwen2.5 models for local LLM work.
- Designed the library interfaces: observe() for the write path and build_context() for the read path, and the data types for turns, memory items and questions.
- Decided to build the evaluation harness before the memory policies, so every later idea is a measured experiment.

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Week No: 4

Date of Reporting: 12/08/2026

08/08/2026 to 13/08/2026

Work carried out

- Prepared evidence-backed answers to the four Review 1 panel questions.
- Moved the contribution from the three-tier architecture to the memory decision policy (keep / compress / forget / recall).
- Built Step 0: evaluation harness, three-tier store, policies P0 (sliding window) and P1 (heuristic scorer), and a synthetic test set.
- 30 automated tests passing; made the P0 baseline budget-driven so the comparison is fair.

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Week No: 5

Date of Reporting: 19/08/2026

15/08/2026 to 20/08/2026

Work carried out

- Found an evaluation leak: a consolidation rule read the answer label (fact_id), worth 45-47 recall points. Fixed it and withdrew the Review 1 figure (95% became 50%).
- Added label-blindness regression tests.
- Wired the LoCoMo benchmark (10 conversations, 1,527 questions) and the MiniLM embedder; added the answer-presence metric.
- Fixed Tier 2 consolidation, improved the compressor, and ran a 23-configuration ablation study.

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Week No: 6

Date of Reporting: 26/08/2026

22/08/2026 to 27/08/2026

Work carried out

- Imposed a storage budget on the whole memory store, not only the prompt.
- Key finding: scored forgetting beats oldest-first forgetting at the same storage, while compressing evicted turns loses answer recall.
- Scaling study to 182x compression; byte accounting including the embedding index.
- Clustered bootstrap confidence intervals and McNemar tests; learned scorer (leave-one-conversation-out); first publication figures; built the P2 LLM judge.

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Week No: 7

Date of Reporting: 02/09/2026

29/08/2026 to 03/09/2026

Work carried out

- Ran the P2 LLM judge (Qwen2.5-0.5B) on GPU, reading salience from the digit-token logits of one forward pass.
- Integrated a fixed reader model (Qwen2.5-1.5B, 4-bit) for end-task accuracy and ran a pilot.
- Pinned every model to a local copy so runs cannot fetch new weights.

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Week No: 8

Date of Reporting: 09/09/2026

05/09/2026 to 10/09/2026

Work carried out

- Measured the P2 judge: it loses to the hand-written heuristic by 5.87 points, while a learned scorer gains 11.47.
- Prepared the Review 2 progress report, a generated deck, a results dashboard and a speaker's guide.
- Wrote the first draft of the IEEE conference paper with tables generated from the result files.

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Week No: 9

Date of Reporting: 16/09/2026

12/09/2026 to 17/09/2026

Work carried out

- Ran end-task accuracy on all 715 answer-recoverable LoCoMo questions.
- Result: +6.0 of the +8.7 recall points become correct answers (69% of the gap), against a 1.7% closed-book floor.
- Updated the report and paper with the end-task results.

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Week No: 10

Date of Reporting: 23/09/2026

19/09/2026 to 24/09/2026

Work carried out

- Added a LongMemEval loader and set up a sweep of store budgets to test the headline on a second benchmark.
- Matched LongMemEval's retention to LoCoMo's so the two are comparable.
- Continued revising the paper draft.

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Week No: 11

Date of Reporting: 30/09/2026

26/09/2026 to 01/10/2026

Work carried out

- LongMemEval result: the headline does not replicate; traced to the heuristic scorer, whose separation of evidence turns reverses sign on that corpus.
- Ruled out the recency-reservation confound.
- Ran abstractive compression and online learning of the scorer at full scale.
- Compiled the paper and cut it to 6 pages; rebuilt the Review 2 deck in the NMIMS format with speaker notes; built the first live demo.

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Week No: 12

Date of Reporting: 07/10/2026

03/10/2026 to 08/10/2026

Work carried out

- Presented Review 2 (midterm review): progress against the plan, design, results so far and the live demo.
- Recorded the panel's feedback and planned the remaining work: a working application and a learned scorer.

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Week No: 13

Date of Reporting: 14/10/2026

10/10/2026 to 15/10/2026

Work carried out

- Turned the demo into a working chat app: saved chats, sidebar, memory panel showing every decision.
- Found and fixed a scorer defect (speaker tag scored with the message) and re-ran the results (P1-S 36.8% to 37.5%).
- Trained the learned scorer on LoCoMo and REALTALK (human chats) and shipped it in the app.
- Updated the report, paper, deck and dashboard to the fixed numbers.

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Week No: 14

Date of Reporting: 21/10/2026

17/10/2026 to 22/10/2026

Work carried out

- Full app: one-command setup, any text input (pastes, PDF, Word, PowerPoint, other languages), a Compare tab and score explanations.
- Stress tests (six input bugs fixed) and a load test with 200 simultaneous users.
- Compared with MemGPT-style and Mem0-style memory: the learned scorer wins at every budget.
- Retrained the scorer on 14.6x the data; chats up to 10M tokens; on million-token chats the app keeps the answer 51.9% of the time vs 19.2% for RAG.
- Google Gemini replies with a local Qwen backup; entry gate and score tags on every message.

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Week No: 15

Date of Reporting: 28/10/2026

24/10/2026 to 29/10/2026

Work carried out

- Prepared the journal version of the paper with Dr. Praveen Modi as co-author and corresponding author.
- Matched the paper to the code; extended references to 52 sources.
- Applied the mentor's review of the paper; rewrote the abstracts and the prose.
- Prepared the log book, project report (black book) and annexure forms.

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Week No: 16

Date of Reporting: 04/11/2026

31/10/2026 to 05/11/2026

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Week No: 17

Date of Reporting: 11/11/2026

07/11/2026 to 12/11/2026

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